

Haloalkanes and Haloarenes

NCERT Class XII Chemistry · Chapter 6 · Haloalkanes and Haloarenes

Haloalkanes and Haloarenes — Complete Notes

6 marks, whole chapter 6.1–6.8, and your first organic chapter of the paper. Organic chemistry runs on a small set of ideas repeated everywhere — nucleophile, leaving group, carbocation, steric hindrance. Every one of them is tagged [exposure] and explained from scratch here, because once you have those four, most of this chapter answers itself.

What these compounds are — *start here — one sentence defines the whole chapter*

[exposure] A **haloalkane** (also called an **alkyl halide**, written $R-X$) is what you get when you take a hydrocarbon chain and replace one or more hydrogen atoms with a halogen — fluorine, chlorine, bromine or iodine. That's the entire definition. Ethane CH_3CH_3 with one hydrogen swapped for chlorine becomes chloroethane CH_3CH_2Cl . The defining structural feature: the carbon carrying the halogen is sp^3 **hybridised** — it has four single bonds, no double bond anywhere on it.

[exposure] A **haloarene** (or **aryl halide**, $Ar-X$) is the same swap performed on an *aromatic* ring instead. Benzene with one hydrogen replaced by chlorine is chlorobenzene. The defining feature here: the carbon carrying the halogen is sp^2 **hybridised**, because it is part of the ring's double-bond system. This single difference — sp^3 versus sp^2 — is why haloalkanes and haloarenes behave almost oppositely, and it comes up in the exam again and again.

Throughout the chapter, X means any halogen and R means any alkyl group.

Why the chapter exists at all

Halogen-containing organic compounds are everywhere, natural and synthetic:

- **Chloramphenicol** — a chlorine-containing antibiotic made by soil microorganisms, used against typhoid.
- **Thyroxine** — the iodine-containing hormone your own thyroid produces. Too little iodine in the diet and you get **goitre**.
- **Chloroquine** — synthetic, used against malaria.
- **Halothane** — a synthetic anaesthetic used during surgery.
- **Fully fluorinated compounds** — every hydrogen replaced by fluorine. They dissolve and deliver oxygen to tissue, so they are being explored as **blood substitutes** in surgery.

Examiner asks: one-mark recall of a named compound and its use. Chloramphenicol/typhoid and thyroxine/goitre are the two that appear.

6.1 Classification — *three separate cuts — count, hybridisation, and where the halogens sit*

By number of halogen atoms

Mono, di or poly halogen compounds, depending on whether they carry one, two, or more halogen atoms. CH_3Cl is monohalo; CH_2Cl_2 is dihalo; CHCl_3 is trihalo. The same labelling works on rings: one chlorine on benzene is a monohaloarene, two is a dihaloarene.

By hybridisation of the carbon holding the halogen

This is the classification that actually predicts behaviour, so learn these five names properly.

Compounds with an sp^3 C-X bond:

[exposure] Alkyl halides — the halogen sits on an ordinary sp^3 carbon of a chain. They are sub-classified by *what else* that carbon is attached to: if the carbon carries one other carbon it is **primary (1°)**, two others **secondary (2°)**, three others **tertiary (3°)**. This count matters more than anything else in the chapter — it decides which mechanism the compound follows and how fast. $\text{CH}_3\text{CH}_2\text{Cl}$ is primary; $(\text{CH}_3)_2\text{CHCl}$ is secondary; $(\text{CH}_3)_3\text{CCl}$ is tertiary.

[exposure] An allylic halide has the halogen on an sp^3 carbon that sits *next to* a carbon-carbon double bond. The word exists because that neighbouring double bond changes everything: when the halogen leaves, the positive charge left behind can spread into the double bond by resonance, so allylic halides react far faster than ordinary ones. Example: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$ — the chlorine-bearing carbon has four single bonds, the carbon beside it has the double bond.

[exposure] A benzylic halide is the same idea with a benzene ring instead of a double bond: the halogen is on an sp^3 carbon directly attached to an aromatic ring. $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ (benzyl chloride) is the standard one. Benzylic halides are the *most* reactive of all in this respect, because the ring gives the intermediate the largest resonance stabilisation available. These too can be 1° , 2° or 3° by the same counting rule.

Compounds with an sp^2 C-X bond:

[exposure] A vinylic halide has the halogen bonded *directly to* a carbon of a carbon-carbon double bond. Not next to it — on it. $\text{CH}_2=\text{CH}-\text{Cl}$ is vinyl chloride. Because that carbon is sp^2 , the C-X bond is shorter and stronger, and these are very unreactive towards substitution.

[exposure] An aryl halide has the halogen bonded directly to an sp^2 carbon of an aromatic ring — chlorobenzene. Same story as vinylic, for the same reason plus resonance.

Trap: allylic/benzylic (fast) versus vinylic/aryl (slow) is decided by *one* question — is the halogen **on** the unsaturated carbon, or **next to** it? On it $\rightarrow sp^2 \rightarrow$ unreactive. Next to it $\rightarrow sp^3 \rightarrow$ very reactive. This exact discrimination appeared as 2017 Q5(b).

Dihaloalkanes: where the two halogens sit

[exposure] Geminal (gem) dihalides carry both halogens on the *same* carbon — from Latin *geminus*, twin. CH_3CHCl_2 is one. Their common name is built from the alkylidene group: CH_3CHCl_2 is **ethylidene chloride**.

[exposure] Vicinal (vic) dihalides carry the halogens on *adjacent* carbons — from Latin *vicinus*, neighbour. $\text{CH}_2\text{ClCH}_2\text{Cl}$ is one. Their common name uses the alkylene group: **ethylene**

dichloride. Vicinal dihalides matter because that's what you get when bromine adds across a double bond.

Examiner asks: "write the structures of all dihalogen derivatives of propane" (NCERT intext 6.3) — there are four: 1,1- and 2,2- (both geminal), 1,2- (vicinal), and 1,3-.

6.2 Nomenclature — IUPAC and common names — both are examinable

IUPAC rules, in the order you apply them

1. Find the **longest carbon chain** containing the principal functional group. Count carbons only — hydrogens don't matter.
2. The halogen is a **substituent**, named as a prefix: **fluoro-, chloro-, bromo-, iodo-**.
3. If there is a **double or triple bond**, it takes **priority** for the lowest number, not the halogen.
4. With no multiple bond, number so the **substituent** gets the lowest locant.
5. If numbering from both ends gives the same set of numbers, give the lowest number to whichever substituent comes **first alphabetically**. Bromo beats methyl; chloro beats methyl.
6. Write substituents alphabetically in the name.

Structure	IUPAC name	Common name
$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	1-bromopropane	<i>n</i> -propyl bromide
$(\text{CH}_3)_2\text{CHCl}$	2-chloropropane	isopropyl chloride
$\text{CH}_3\text{CH}_2\text{CHClCH}_3$	2-chlorobutane	<i>sec</i> -butyl chloride
$(\text{CH}_3)_3\text{CBr}$	2-bromo-2-methylpropane	<i>tert</i> -butyl bromide
$(\text{CH}_3)_2\text{CHCH}_2\text{Br}$	1-bromo-2-methylpropane	isobutyl bromide
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$	1-chlorobutane	<i>n</i> -butyl chloride

The straight-chain compound always takes ***n*-** in the common name. A three-carbon group halogenated at the middle carbon is **isopropyl**. Four carbons branched at the end is **isobutyl**; branched at the halogen-bearing carbon is *sec*- or *tert*-butyl depending on the count.

Common names you're expected to know cold

Formula	IUPAC	Common
CHCl_3	trichloromethane	chloroform
CHBr_3	tribromomethane	bromoform
CHI_3	triiodomethane	iodoform
CCl_4	tetrachloromethane	carbon tetrachloride
CH_2Cl_2	dichloromethane	methylene chloride

Aromatic compounds

For haloarenes the common name and the IUPAC name are the same word — chlorobenzene is both. Three side-chain names you must recognise:

- **Benzyl** — benzene ring plus CH_2 . So $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ is benzyl chloride.
- **Benzal** — ring plus CH carrying two halogens. $\text{C}_6\text{H}_5\text{CHCl}_2$ is benzal dichloride.
- **Benzo-** — ring plus a carbon with no hydrogen left. $\text{C}_6\text{H}_5\text{CCl}_3$ is benzo-trichloride.

For two substituents on a ring, positions 1,2 are **ortho (o-)**, 1,3 are **meta (m-)**, 1,4 are **para (p-)**. So 1,4-dibromobenzene is *p*-dibromobenzene. Three substituents at 1,3,5 is **sym-** (symmetrical).

Examiner asks: "give the IUPAC name of the following" is a standing 3-marker (2017 Q14, 2024 Q11a). Always check for a double bond before you number.

6.3 Nature of the C-X bond — *short section, but it explains every reaction that follows*

Halogens are more electronegative than carbon, so the shared electron pair sits closer to the halogen. The result is a **polar bond**: the carbon carries a partial positive charge $\delta+$ and the halogen a partial negative charge $\delta-$.



That $\delta+$ carbon is the target of essentially every reaction in this chapter. Anything electron-rich will go for it.

Down the group $\text{F} \rightarrow \text{Cl} \rightarrow \text{Br} \rightarrow \text{I}$, the halogen gets bigger, so the **bond length increases** and the **bond enthalpy decreases**. A longer, weaker bond breaks more easily — which is why R-I is the most reactive alkyl halide and R-F the least in substitution reactions. Under ordinary conditions R-F barely reacts at all.

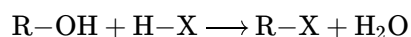
Trap: the C-Cl bond in a haloalkane is 177 pm, but in a haloarene only 169 pm — shorter, therefore stronger, therefore harder to break. That number is the physical reason haloarenes resist nucleophilic substitution, and it is worth quoting.

6.4 Preparation of haloalkanes — *four routes — from alcohols, from alkanes, from alkenes, by exchange*

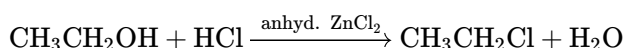
A · From alcohols

The general move: replace the $-\text{OH}$ group with a halogen.

(i) With halogen acids (HX)

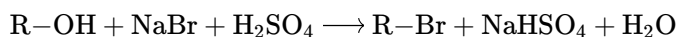


For **chloroalkanes**, use HCl with **anhydrous** ZnCl_2 as catalyst:



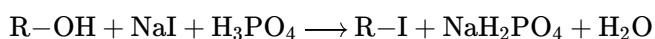
The ZnCl_2 works as a **dehydrating agent**: it soaks up the water product. By Le Chatelier's principle, removing a product drives the equilibrium forward, so you get more chloroethane.

For **bromoalkanes**, heat the alcohol with NaBr (or KBr) and concentrated H_2SO_4 :



[exposure] The HBr is generated **in situ** — Latin for "in place", meaning the reagent is made inside the reaction vessel at the moment it's needed rather than being bought and stored. Chemists do this whenever a reagent is nasty or unstable to keep around. HBr is a corrosive acid that gives off heavy fumes, so making it in the flask from $\text{NaBr} + \text{H}_2\text{SO}_4$ is simply safer. You can also just use 48% concentrated HBr directly.

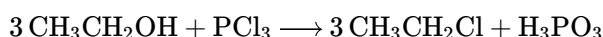
For **iodoalkanes**, use NaI (or KI) with **95% phosphoric acid**, H_3PO_4 — *not* sulphuric acid:



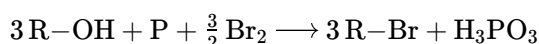
Trap — this is NCERT intext 6.2 and it appears in boards: why not H_2SO_4 with KI? Concentrated H_2SO_4 *would* make HI, but it is also a **strong oxidising agent** and immediately oxidises that HI to I_2 . The HI never survives to reach the alcohol. H_3PO_4 is not an oxidising agent, so it works.

Reactivity orders to memorise — halogen acid: $\text{HI} > \text{HBr} > \text{HCl}$ (because HI has the longest, weakest bond, so it ionises most easily). Alcohol: **tertiary** > **secondary** > **primary** (because these reactions go through a carbocation, and a tertiary carbocation is the most stable).

(ii) With phosphorus halides



For bromides and iodides, use **red phosphorus with** Br_2 **or** I_2 . The PBr_3 or PI_3 is generated in situ, again because these are too unstable to store:



(iii) With thionyl chloride — the preferred method

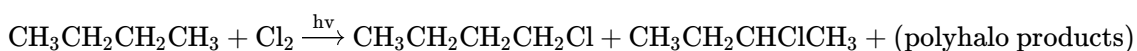


Examiner asks: "why is thionyl chloride preferred for preparing alkyl chlorides from alcohols?" (2024 Q11b) — because both by-products, SO_2 and HCl , are **gases that escape**, leaving pure alkyl chloride behind with no separation step. Say "escapable gases, hence pure product" and you have the mark.

Trap: none of these routes work on phenol. In phenol, the oxygen's lone pair delocalises into the ring, giving the C-O bond **partial double-bond character**. It's too strong to break, so you cannot make an aryl halide from phenol this way. This is a stock 2-marker and it also shows up embedded in "major monohalo product" questions where a molecule has two –OH groups: only the non-phenolic one reacts.

B • From hydrocarbons — free radical halogenation

Treat an alkane with Cl₂ or Br₂ in the presence of UV light or heat, and hydrogens get replaced one at a time:



The problem: you get a **complex mixture** of isomeric mono- and polyhalogenated products whose boiling points are so close they are almost impossible to separate. So this is **not a preferred preparative method** — but it is heavily examined for *predicting* products.

When asked for the **major** free-radical product, use the free-radical stability order: **benzylic** $\approx$ **allylic** > **tertiary** > **secondary** > **primary**. Benzylic and allylic radicals win because the ring or double bond stabilises them by resonance.

C • From alkenes

(i) Addition of a halogen acid

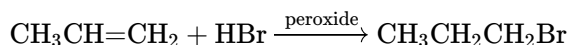
With a **symmetrical** alkene, there's no choice to make:



[exposure] With an **unsymmetrical** alkene the two carbons of the double bond are different, so you need a rule to say which end the halogen goes to. **Markovnikov's rule** says the **negative part of the adding molecule attaches to the carbon that carries fewer hydrogens**. The reason underneath is carbocation stability — the proton adds first, wherever it produces the more stable carbocation, and the halide then goes to that carbon. So propene plus HBr gives 2-bromopropane, not 1-bromopropane.

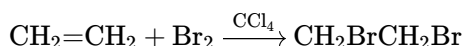


Anti-Markovnikov (peroxide / Kharasch effect): add a peroxide such as benzoyl peroxide and the rule flips — bromine goes to the carbon with *more* hydrogens.



Trap: the peroxide effect works with HBr **only**. It does not happen with HCl or HI. If a question gives you peroxide with HCl, the answer is still Markovnikov.

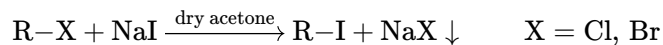
(ii) Addition of a halogen



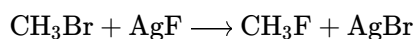
This gives a **vicinal dibromide**. The reddish-brown colour of the bromine solution **discharges** as the reaction proceeds — which is exactly the standard laboratory **test for unsaturation**.

D • Halogen exchange

[exposure] The **Finkelstein reaction** makes alkyl *iodides* by swapping out a chloride or bromide, using NaI in **dry acetone**. It needs a named trick because the swap is an equilibrium that wouldn't otherwise favour the product. The trick: NaCl and NaBr are **insoluble in dry acetone and precipitate out**, while NaI is soluble. Removing a product drags the equilibrium forward (Le Chatelier again).



[exposure] The **Swarts reaction** makes alkyl *fluorides*, which the ordinary routes can't touch. Heat an alkyl chloride or bromide with a **metallic fluoride** — AgF, Hg₂F₂, CoF₂ or SbF₃:

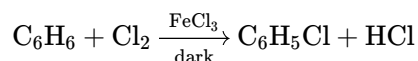


Examiner asks: "what happens when methyl bromide is treated with silver fluoride?" (2025 Q20a) — name it as Swarts and give the equation.

6.5 Preparation of haloarenes — *two routes — and one of them is a named reaction you must know*

A • From arenes, by electrophilic substitution

Treat benzene with Cl₂ or Br₂ in the presence of a **Lewis acid catalyst** — FeCl₃, FeBr₃ or iron filings — in the dark:



Mechanically: the Lewis acid is electron-deficient, so it pulls Cl[−] off Cl₂, leaving Cl⁺. That Cl⁺ is the **electrophile** that attacks the electron-rich ring.

With **toluene**, the −CH₃ group is ortho/para directing, so you get a mixture of *o*- and *p*-chlorotoluene. Conveniently, these two are **easy to separate** because their melting points differ widely.

Two limits on this method:

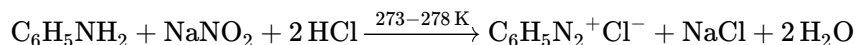
- **Fluoro compounds can't be made this way** — fluorine is too reactive and the reaction is uncontrollable.
- **Iodination is reversible** and gives poor yield. Fix it by adding an **oxidising agent** (HNO₃ or HIO₄), which oxidises the HI by-product to I₂ and so pulls the equilibrium forward.

B • From amines — the Sandmeyer reaction

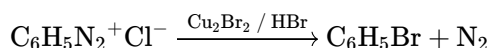
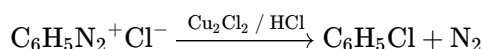
[exposure] A **diazonium salt** is an aromatic compound carrying the group −N₂⁺ — a nitrogen pair on its way out. It exists as an intermediate because that group is an *outstanding*

leaving group: it departs as harmless N₂ gas, dragging the reaction forward. That makes it the standard way to put almost anything onto a benzene ring. Diazonium salts are only stable cold, which is why the next step specifies 0–5 °C.

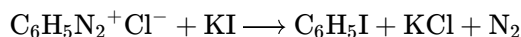
Step 1 — diazotisation. Treat a primary aromatic amine (aniline) with nitrous acid, generated in situ from NaNO₂ + HCl, at **273–278 K (0–5 °C)**:



Step 2 — Sandmeyer. Treat the diazonium salt with a **cuprous halide**:



Trap: for the **iodo** compound you need **no copper salt at all** — simply shaking the diazonium salt with KI does it. This exception is asked directly.



6.6 Physical properties — *every point here is a "give the reason" question*

Physical state and colour

CH₃Cl, CH₃Br, CH₃CH₂Cl and some chlorofluoromethanes are **gases** at room temperature; higher members are liquids or solids. Pure haloalkanes are **colourless**, but bromo- and iodo-compounds develop colour on standing in light.

Boiling point — four separate rules

1. RF < RCl < RBr < RI. Down the group the halogen's mass and size increase, so the **van der Waals forces** between molecules get stronger and more energy is needed to separate them.
2. **A haloalkane always boils higher than the hydrocarbon of comparable mass.** Two reasons together: the C–X bond is polar, so molecules attract each other by **dipole-dipole interaction**; and the heavy halogen raises the molecular mass, strengthening van der Waals forces.
3. **Among isomers, boiling point falls as branching rises.** Branching makes a molecule more compact and spherical, cutting the **surface area of contact** between molecules, which weakens van der Waals forces.
4. **More halogen atoms → higher boiling point**, again through mass.
CH₃Cl < CH₃Br < CH₂Br₂ < CHBr₃.

Worked: Arrange in increasing boiling point: 1-chloropropane, 2-chloropropane, 1-chlorobutane. 1-chlorobutane has four carbons — highest mass, so it boils highest. The other

two are isomers, so compare branching: 2-chloropropane is branched, so it boils lowest.

Answer: **2-chloropropane < 1-chloropropane < 1-chlorobutane.**

Melting point — the para exception

The boiling points of *o*-, *m*- and *p*-dihalobenzenes are nearly identical. But the **para isomer melts much higher** than the other two, because it is more **symmetrical** and so packs more tightly into the crystal lattice. More energy is needed to break a well-packed lattice apart.

Density

Bromo, iodo and polychloro derivatives are **heavier than water**. Density increases with the number of carbon atoms, the number of halogen atoms, and the atomic mass of the halogen.

Solubility — a favourite "give the reason"

Haloalkanes are polar, yet they are **insoluble in water**. Dissolving anything in water requires breaking the strong **hydrogen bonds between water molecules** and also the dipole-dipole attractions between haloalkane molecules. The energy released when new haloalkane-water attractions form is **less** than the energy needed to break the old ones — so it doesn't happen.

They *are* soluble in organic solvents (alcohol, ether, benzene), because there the new attractions formed are about as strong as the ones broken, so the books balance.

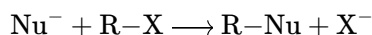
Examiner asks: "why are alkyl halides, though polar, immiscible with water?" (2017 Q5a) — the two energies, in one sentence each. It is a repeat question.

6.7 Reactions of haloalkanes — nucleophilic substitution — *the core of the chapter; get the two words first*

[exposure] A **nucleophile** is a species that is **electron-rich** — it has a lone pair or a negative charge — and goes looking for a positive centre to donate that pair to. The name means "nucleus-loving". OH^- , CN^- , NH_3 and RO^- are all nucleophiles. In this chapter the positive centre it attacks is always the $\delta+$ carbon of the C-X bond.

[exposure] A **leaving group** is the species that departs, taking the bonding electron pair with it. Here it's the halide ion X^- . A group leaves easily when the resulting ion is stable and the bond was weak — so I^- **is the best leaving group and F^- the worst**, following bond strength. That single fact explains the reactivity order $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$.

Put those together and the reaction reads: a nucleophile attacks the $\delta+$ carbon, the halide leaves, and you have a new compound. The substitution only happens if the incoming nucleophile is **stronger than the halide it displaces**.



The substitution products you must be able to write

Reagent	Product	Class of product
aqueous KOH / NaOH	R–OH	alcohol
NaOR (sodium alkoxide)	R–O–R	ether
KCN	R–CN	alkyl cyanide (nitrile)
AgCN	R–NC	alkyl isocyanide
NH ₃	R–NH ₂	primary amine
KNO ₂	R–O–N=O	alkyl nitrite
AgNO ₂	R–NO ₂	nitroalkane
R'COOAg (silver carboxylate)	R'COOR	ester
LiAlH ₄	R–H	alkane (same carbon count)
R'–Li (or Na)	R–R'	alkane (carbons added)

[exposure] Look at the four pairs above that come in twos. An **ambident nucleophile** is a nucleophile with **two different donor atoms**, either of which can do the attacking — though only one at a time. The category exists because *which* atom attacks decides which product you get, and that in turn depends on whether the reagent is ionic or covalent. The two examples are CN[–] (attacks through C or N) and NO₂[–] (attacks through N or O).

Trap — and one of the most-asked questions in the chapter: KCN is **ionic**, so it dissociates and gives a free CN[–] ion which attacks through its **carbon** → alkyl **cyanide**. AgCN is **covalent**, so it does not dissociate; only the **nitrogen** lone pair is available → alkyl **isocyanide**. The same logic runs the KNO₂ / AgNO₂ pair: ionic gives attack through oxygen (nitrite ester), covalent gives attack through nitrogen (nitroalkane). Remember it as "**silver goes through nitrogen**".

Mechanism 1 • S_N2 — bimolecular

[exposure] S_N2 stands for **Substitution, Nucleophilic, bimolecular**. "Bimolecular" means **two** species are involved in the slow, rate-deciding step, so the rate depends on the concentration of *both*:

$$\text{Rate} = k [\text{R–X}][\text{Nu}^-]$$

Everything happens in **one single step**. The nucleophile attacks the carbon from the side **opposite** the halogen — it has to, because the halogen's lone pairs would repel it if it came from the front. As the new C–Nu bond forms, the old C–X bond breaks, **simultaneously**.

[exposure] Halfway through, the carbon is momentarily bonded to **five** things at once — the three groups it keeps, the incoming nucleophile, and the leaving halogen. That arrangement is the **transition state**. It is not an intermediate: it is a fleeting, very high-energy structure that **cannot be isolated**, existing only at the top of the energy hill. In S_N2 there is **no intermediate at all** — no carbocation, no free radical.

Because the nucleophile comes in from the back and the halogen leaves the front, the other three groups get pushed through, like an umbrella flipping in the wind. The result is

inversion of configuration, called Walden inversion.

Worked: (–)-2-bromooctane treated with OH^- gives (+)-octan-2-ol. The OH ends up in the position the Br did not occupy — the configuration is inverted, and the sign of optical rotation flips.

[exposure] Steric hindrance is physical crowding: bulky groups around the reaction centre get in the way of an incoming reagent and slow the reaction down. It matters here because the $\text{S}_{\text{N}}2$ nucleophile must physically reach the carbon. A methyl halide has only three small hydrogens in the way, so it reacts fastest. A tertiary halide has three fat CH_3 groups blocking the approach, so it reacts slowest.

$\text{S}_{\text{N}}2$ **reactivity:** $\text{CH}_3\text{X} > \text{primary} > \text{secondary} > \text{tertiary}$.

Trap: when both compounds in a pair are primary, compare **how close the branching is to the halogen**. A methyl group nearer the C–X carbon crowds it more and slows $\text{S}_{\text{N}}2$ further. That comparison is NCERT intext 6.7 and appears in boards.

Mechanism 2 • $\text{S}_{\text{N}}1$ — unimolecular

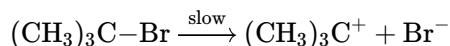
[exposure] $\text{S}_{\text{N}}1$ is **Substitution, Nucleophilic, unimolecular**. "Unimolecular" means only **one** species appears in the slow step — the alkyl halide. So the rate depends only on *its* concentration; changing the nucleophile's concentration does nothing at all:

$$\text{Rate} = k [\text{R-X}]$$

It runs in **two steps**, and it requires a **polar protic solvent** — polar (it has $\delta+$ / $\delta-$ ends) and protic (it contains an O–H or N–H that can hand out protons). Water, alcohols and acetic acid all qualify.

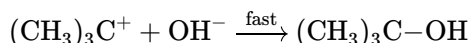
[exposure] A **carbocation** is a carbon atom bearing a **full positive charge**, left behind when a leaving group departs with both bonding electrons. It has only three bonds, so it is sp^2 hybridised and **flat (planar)**. Being electron-deficient it is very unstable and short-lived — but its stability varies enormously, and that variation drives this whole mechanism. **Tertiary > secondary > primary > methyl**, because the surrounding alkyl groups push electron density in and cushion the positive charge. Benzylic and allylic carbocations beat all of these, because resonance spreads the charge over several atoms.

Step 1 (slow, reversible — the rate-determining step):



Breaking that C–Br bond costs energy. The energy comes from **solvation**: the protic solvent's $\delta+$ hydrogens crowd around the departing Br^- , stabilise it, and release energy in doing so. That is exactly why a polar protic solvent is required.

Step 2 (fast): the nucleophile attacks the carbocation.



$\text{S}_{\text{N}}1$ **reactivity:** **tertiary > secondary > primary** — the exact opposite of $\text{S}_{\text{N}}2$, and for a completely different reason (carbocation stability, not crowding).

Trap: for allylic and benzylic halides, **both** mechanisms are fast — S_N1 because their carbocations are resonance-stabilised, S_N2 because the carbon is usually primary and uncrowded. Don't let "it's primary" push you to the wrong answer on an S_N1 question about benzyl chloride: benzyl chloride is the *faster* S_N1 substrate against cyclohexyl chloride, because the benzyl carbocation is resonance-stabilised.

S_N1 vs S_N2 — the comparison table

	S_N1	S_N2
Order of reaction	first order	second order
Rate depends on	[RX] only	[RX] and [Nu]
Steps	two	one
Goes through	carbocation intermediate	transition state, no intermediate
Stereochemistry	racemisation	inversion
Favoured by	$3^\circ > 2^\circ > 1^\circ$	$1^\circ > 2^\circ > 3^\circ$
Solvent	polar protic	polar aprotic

Examiner asks: "write two differences between S_N1 and S_N2 " (2021 Q21) — pick order + steps, or intermediate + stereochemistry. Two rows, two marks.

Optical isomerism — *the stereochemistry vocabulary* — *six terms, each needed for the S_N answers*

[exposure] Optical activity is a property some substances have of **rotating the plane of plane-polarised light**. Ordinary light vibrates in every direction; pass it through a polariser and it vibrates in one plane only. Send that through certain liquids or solutions and the plane comes out twisted by some angle. Substances that do this are **optically active**; those that don't are optically inactive. The angle is measured with an instrument called a **polarimeter**.

A substance rotating the plane **clockwise (right)** is **dextrorotatory**, written **d** or **(+)**. One rotating it **anticlockwise (left)** is **laevorotatory**, written **l** or **(-)**.

[exposure] A chiral carbon (also called an asymmetric carbon or stereocentre) is an sp^3 carbon attached to **four different atoms or groups**. It is marked with an asterisk. It matters because it is the structural cause of optical activity — with four different groups the molecule has no plane of symmetry, and that asymmetry is what twists the light. In $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$ the second carbon holds methyl, ethyl, bromine and hydrogen: four different things, so it is chiral.

[exposure] A chiral molecule has no plane of symmetry, and so is **non-superimposable on its mirror image** — exactly like your left and right hands, which is where the name comes from (Greek *cheir*, hand). Lay one over the other and they never match. A molecule with a chiral carbon is normally chiral, and chiral molecules are optically active.

[exposure] Enantiomers are the two mirror-image forms of a chiral molecule. They are stereoisomers — same connections, different spatial arrangement. They rotate plane-polarised light by the **same angle in opposite directions**: if one is dextro, the other is laevo. Every

other physical property — melting point, boiling point, solubility, refractive index — is **identical**.

Examiner asks: "why are the d- and l-isomers of butan-2-ol difficult to separate by fractional distillation?" (2024 Q22) — because fractional distillation separates on boiling point, and enantiomers have **identical** boiling points. One line, full marks.

[exposure] A **racemic mixture** is a **1:1 (equimolar)** mixture of the two enantiomers. It is **optically inactive**, because the rotation produced by one is exactly cancelled by the other. It's written **dl** or **(±)**, as in (±)-butan-2-ol. The process of converting a single enantiomer into such a mixture is **racemisation**.

[exposure] Three words for what a substitution does to a chiral centre. **Retention** — the new group takes the same spatial position the old one had; the configuration is unchanged.

Inversion — the new group takes the position *opposite*; the configuration is flipped.

Racemisation — you get roughly 50% of each, so the product is optically inactive.

Which mechanism gives which:

- $S_N2 \rightarrow$ **inversion**. The nucleophile can only attack from the back, so there is only one possible outcome.
- $S_N1 \rightarrow$ **racemisation**. The carbocation intermediate is **planar**, so the nucleophile can attack from either face with equal ease. Two products form in equal amounts, they are enantiomers, and the mixture is racemic.

Examiner asks: "out of S_N1 and S_N2 , which occurs with inversion and which with racemisation?" (2026 Q23c) — and expect to have to *explain* it: back-side attack for one, planar carbocation for the other.

Elimination and reactions with metals — *the other two things a haloalkane does*

Elimination — dehydrohalogenation

[exposure] **Dehydrohalogenation** means removing H and X from adjacent carbons to make a double bond. It is also called **β-elimination**, because of where the two pieces come from: the carbon bearing the halogen is the **α-carbon**, the one next to it is the **β-carbon**, and the reaction takes the halogen from α and a hydrogen from β. The reagent is **alcoholic KOH** — and the "alcoholic" is the whole point, because aqueous KOH gives substitution instead.



Trap — the single most reliable trick in the chapter: aqueous KOH → alcohol (substitution). Alcoholic KOH → alkene (elimination). One word in the question decides the whole answer, and it is tested constantly.

[exposure] When a molecule has **more than one β-hydrogen**, more than one alkene can form, and **Saytzeff's rule** (NCERT spells it **Zaitsev**, "also pronounced as Saytzeff") tells you which one dominates: the **more substituted alkene** — the one whose double-bonded carbons

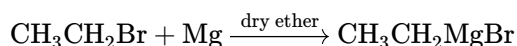
carry the most alkyl groups — is the major product. It exists as a rule because more-substituted alkenes are more stable, and the reaction settles into the more stable product.

Worked: 2-bromopentane with alcoholic KOH. There are β -hydrogens on both sides, so two alkenes are possible. Removing one gives pent-1-ene, the other gives pent-2-ene. Pent-2-ene has an alkyl group on *each* double-bonded carbon, pent-1-ene has only one — so **pent-2-ene is major, 81% against 19%**.

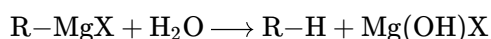
Reaction with metals

[exposure] An **organometallic compound** is any compound containing a direct **carbon-metal bond**. They matter because the metal, being electropositive, pushes electron density onto the carbon, giving that carbon a δ^- charge — the exact reverse of its usual state. A carbon that is normally attacked becomes a carbon that attacks.

[exposure] The **Grignard reagent**, $R-Mg-X$ (alkyl magnesium halide), is the most important organometallic compound in the syllabus. Make it by reacting a haloalkane with **magnesium metal in dry ether**:

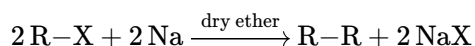


The C-Mg bond is covalent but **highly polar**: carbon δ^- , magnesium δ^+ . That δ^- carbon is so electron-rich that it grabs a proton from **any** source — water, alcohol, an amine — and turns into an alkane:



Examiner asks: "why should Grignard reagents be prepared under anhydrous conditions?" (2024 Q13c) — because they react instantly with any water present to give the alkane, destroying the reagent before it can be used. Repeat question; know it verbatim.

[exposure] The **Wurtz reaction** joins two alkyl halides end to end using **sodium in dry ether**, producing an alkane with **double the carbon count**. It exists as a named reaction because it's the standard way to build a longer chain, and because it's the only route that *doubles* — which is what lets you work backwards from the product in exam questions.



Trap: Wurtz only gives a clean product with a **symmetrical** alkane. So when a question gives you the product and asks for the starting halide, **cut the product exactly in half** and put X on the cut. 2,5-dimethylhexane cuts into two 1-bromo-2-methylpropane molecules. That reverse-engineering step is asked repeatedly (2026 Q10b).

Reactions of haloarenes — *why they behave backwards, and the two named ring reactions*

Why haloarenes resist nucleophilic substitution

This is a guaranteed question. There are four reasons; two or three earn full marks.

1. **Resonance.** The halogen's lone pair delocalises into the ring, so the C-X bond acquires **partial double-bond character**. A partial double bond is far harder to break than a single one.
2. **sp^2 hybridisation.** The carbon holding the halogen is sp^2 , so it has more **s-character** and is more electronegative than an sp^3 carbon. It holds the bonding pair more tightly. The bond is also shorter — 169 pm against 177 pm in a haloalkane.
3. **Instability of the phenyl cation.** An S_N1 route would need a phenyl carbocation, which gets no resonance stabilisation and is therefore not formed.
4. **Electronic repulsion.** The ring's π -electron cloud repels an approaching nucleophile.

The nitro group changes everything

Put a **strongly electron-withdrawing group** such as $-\text{NO}_2$ at the **ortho or para** position and the picture flips. The nitro group pulls electron density away, weakening the C-X bond and activating the ring towards nucleophilic attack. Each extra nitro group makes it easier:

Compound	Conditions needed
chlorobenzene	NaOH, 623 K, 300 atm
<i>p</i> -nitrochlorobenzene	NaOH, 443 K, then acidify
2,4-dinitrochlorobenzene	NaOH, 368 K, then acidify
2,4,6-trinitrochlorobenzene	warm water alone → picric acid

Trap: a nitro group at the **meta** position has **no effect at all**. Resonance only pushes and pulls electron density at the ortho and para positions — meta is never in the delocalisation path. Stating this earns a mark on its own.

Electrophilic substitution — the halogen's split personality

Halogens are **ortho/para directing but deactivating**. Both halves of that need explaining:

- **Ortho/para directing** because the halogen's lone pair delocalises *into* the ring (**+R effect**), and resonance raises the electron density specifically at the ortho and para positions. An incoming electrophile goes where the electrons are.
- **Deactivating** because the halogen is also strongly electronegative and pulls electron density *out* along the sigma bond (**-I effect**). Overall the ring ends up poorer in electrons than plain benzene, so the reaction is slower and needs harsher conditions.

Examiner asks: "although chlorine shows a strong -I effect, why is it ortho/para directing?" (2026 Q25a) — answer with the two effects working in opposite directions: -I decides *how fast*, +R decides *where*.

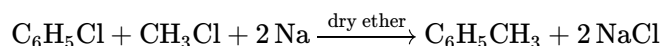
The four standard electrophilic substitutions on chlorobenzene, all giving ortho + para with **para as the major product** (it's more symmetrical, and less crowded than ortho):

- **Halogenation** — Cl_2 , anhydrous $\text{FeCl}_3 \rightarrow$ 1,2- and 1,4-dichlorobenzene.
- **Nitration** — conc. HNO_3 + conc. $\text{H}_2\text{SO}_4 \rightarrow$ 1-chloro-2-nitrobenzene and 1-chloro-4-nitrobenzene.
- **Sulphonation** — conc. H_2SO_4 , heat $\rightarrow$ 2- and 4-chlorobenzenesulphonic acid.

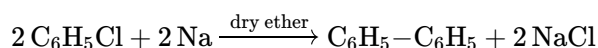
- **Friedel-Crafts**, anhydrous AlCl_3 — **alkylation** with CH_3Cl gives 1-chloro-2/4-methylbenzene; **acylation** with CH_3COCl gives 2- and 4-chloroacetophenone.

Reactions with metals

[exposure] The **Wurtz-Fittig reaction** is the Wurtz reaction run with a **mixture** of an alkyl halide and an aryl halide, using sodium in dry ether. The alkyl group ends up attached to the ring, giving an **alkylarene**. It's the standard way to put a side chain on benzene: chlorobenzene plus CH_3Cl gives **toluene**.



[exposure] The **Fittig reaction** is the same setup with **aryl halide alone** — so two rings couple to each other, giving **diphenyl (biphenyl)**.



Trap: one reagent tells you which. **Both** halides present → Wurtz-Fittig → alkylarene. **Only** the aryl halide → Fittig → biphenyl. "How do you convert chlorobenzene to biphenyl?" (2025 Q12a) is Fittig, not Wurtz-Fittig.

6.8 Polyhalogen compounds — *pure recall — uses and harmful effects, and it's free marks*

[exposure] A **polyhalogen compound** is simply a carbon compound carrying **more than one halogen atom**. The section exists in the syllabus not for the chemistry but for the applications and the environmental damage — so the marks here come from recall, not reasoning.

Dichloromethane, CH_2Cl_2 (**methylene chloride**)

Uses: solvent, paint remover, propellant in aerosols, degreasing agent.

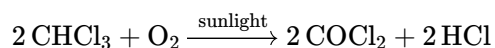
Harm: affects the **central nervous system**. At low levels in air, impairs hearing and vision. At high levels, causes dizziness, nausea, and tingling or numbness in the fingers and toes. Direct contact burns the skin; contact with the eye burns the cornea.

Trichloromethane, CHCl_3 (**chloroform**)

Uses: solvent for fats, alkaloids and iodine; feedstock for making **Freon R-22**. It was once used as a general anaesthetic in surgery but has been replaced by ether, which is less harmful.

Harm: inhaling it **depresses the central nervous system**. Breathing about 900 ppm briefly causes dizziness, fatigue and headache. Chronic exposure damages the **liver and kidneys**.

Examiner asks: "why is chloroform stored in dark coloured bottles?" (2026 Q24a) — because in **light and air it is slowly oxidised to phosgene**, COCl_2 , an extremely poisonous gas. The bottles are dark **and filled to the brim**, so no air is left inside. Repeat question — memorise the equation:



Triiodomethane, CHI_3 (iodoform)

Formerly used as an **antiseptic**, working through the free iodine it liberates. Now replaced by other iodine formulations because of its **objectionable smell**.

Tetrachloromethane, CCl_4 (carbon tetrachloride)

Uses: refrigerant manufacture, aerosol propellant, feedstock for chlorofluorocarbons, solvent. Until the 1960s it was a cleaning fluid — an industrial **degreasing agent** and a household spot remover.

Harm: long exposure causes **liver cancer**. Inhalation causes dizziness, light-headedness, nausea and vomiting, and can permanently damage nerve cells. Severe cases lead to stupor, coma, unconsciousness, irregular heartbeat or death. It also **depletes the ozone layer**.

Freons

Freons are the **chlorofluorocarbons (CFCs)** of methane and ethane. They are extremely stable, unreactive, non-toxic, non-corrosive and easily liquefiable — which is exactly what made them so useful and so damaging. The commonest is **Freon-12**, CCl_2F_2 , made from CCl_4 by the **Swarts reaction**.

Uses: aerosol propellant, refrigeration, air conditioning.

Harm: because they are unreactive, they survive the journey up to the **stratosphere** unchanged. There they start a **chain reaction with ozone** and destroy the ozone layer.

DDT

p,p'-Dichlorodiphenyltrichloroethane — two benzene rings each carrying a chlorine at the para position, joined by a carbon bearing a CCl_3 group. Prepared by the **condensation of chloral with chlorobenzene in sulphuric acid**.

Use: the first modern insecticide — effective against the mosquitoes that spread malaria and the lice that spread typhus.

Harm: many insect species **developed resistance** to it. It is highly toxic to fish. And it is chemically very stable and **fat-soluble**, so animals cannot metabolise it — it accumulates in fatty tissue and builds up along the food chain.

Patterns, collected — *the six things you'll actually be asked to do*

A · Predict the product — 2-3 marks

Recognise it: a reaction arrow with reagents written over it, and "write the major product" or "identify A and B".

1. **Read the reagent first** — it decides everything. Aqueous KOH → alcohol. Alcoholic KOH → alkene. Mg/dry ether → Grignard. Na/dry ether → Wurtz. NaI/acetone → Finkelstein. AgF → Swarts. SOCl_2 → alkyl chloride.

2. If a double bond is being attacked: Markovnikov unless "peroxide" appears, and peroxide only counts with HBr.
3. If an alkene is being formed: apply Saytzeff — more substituted is major.
4. If a ring is being attacked: ortho and para, para major.

B · Which reacts faster, and why — 2-3 marks

Recognise it: a pair of compounds and "which undergoes faster S_N1/S_N2 ".

1. Identify which mechanism the question names. They run in opposite directions.
2. S_N2 → think **steric hindrance**. Fewer/smaller groups around the carbon wins. $1^\circ > 2^\circ > 3^\circ$.
3. S_N1 → think **carbocation stability**. Tertiary, benzylic or allylic wins.
4. If the halogens differ, the **leaving group** decides: $I > Br > Cl > F$.
5. Always give the *reason* word — "less steric hindrance" or "resonance-stabilised carbocation". The reason carries the mark, not the choice.

C · Give the reason — 1-2 marks

Recognise it: "account for the following" or "why is...".

Almost every one of these reduces to one of five stock answers: **resonance / partial double bond** (haloarene unreactivity, phenol won't react, allyl beats propyl), **sp^2 vs sp^3 and s-character** (dipole moment, bond length), **energy of bonds broken vs formed** (water insolubility), **van der Waals and surface area** (boiling points), **escaping gaseous by-products** (thionyl chloride).

D · Arrange in order — 1-2 marks

Recognise it: three or four compounds, "increasing order of...".

1. **Boiling point** → molecular mass first, branching second.
2. **Reactivity to S_N1** → carbocation stability.
3. **Reactivity to S_N2** → steric hindrance.
4. **Reactivity of RX generally** → C-X bond strength: $RI > RBr > RCl > RF$.

E · Conversions — 3 marks

Recognise it: "how would you convert X into Y".

1. Count the carbons on both sides. Same count → substitution or addition. Doubled → Wurtz.
2. Alkene → halide: HX, checking Markovnikov versus peroxide.
3. Need an iodide → make the bromide first, then Finkelstein.
4. Halide → alkene: alcoholic KOH, then check Saytzeff.
5. Halide → alkane: Mg/dry ether then water, or $LiAlH_4$ directly.

F · Work backwards from the product — 2-3 marks

Recognise it: the product is given and the starting material is A, or "identify the hydrocarbon".

1. **Wurtz product** → cut the alkane in half at the middle and add X.
2. **Grignard then water** → the alkane's carbon skeleton is the halide's; put MgX back where the new hydrogen went.

3. "Only one monochlorination product" → all the hydrogens must be equivalent → **2,2-dimethylpropane (neopentane)**.
4. "Three isomeric products" → pentane. "Four" → 2-methylbutane.

Past year questions · question types — *ranked by how often they turn up*

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · Predict the product / identify A, B, C — 3-5 marks

Recognise it: a reaction scheme, sometimes a chain of three or four arrows.

1. Work one arrow at a time; never skip to the end.
2. Name the reaction under each arrow before you draw anything — Finkelstein, Wurtz, Friedel-Crafts, Sandmeyer.
3. Where two products are possible, say which is major and give the rule by name (Markovnikov, Saytzeff, para-major).

Trap: a molecule with two –OH groups, one on the ring and one on a side chain, treated with SOCl_2 or HCl . Only the **side-chain** –OH reacts; the phenolic one has partial double-bond character and stays put.

2 · S_N1 / S_N2 comparison — 2-3 marks

Recognise it: a pair of halides, or "write the mechanism of the following".

1. For a mechanism question, draw **both steps** of S_N1 and label the slow step as rate-determining, or draw the S_N2 transition state in square brackets.
2. For a comparison, name the deciding factor before you name the winner.

Trap: S_N2 ordering is $1^\circ > 2^\circ > 3^\circ$; S_N1 is the exact reverse. Writing the reactivity order without checking which mechanism the question asked about is the commonest way to lose these marks.

3 · "Give the reason" / "account for the following" — 1-2 marks

Recognise it: a statement of fact, then "why?".

1. Identify which of the five stock explanations applies (see Patterns, C).
2. Write it as cause → effect in one or two sentences. No padding.

4 · IUPAC naming and structure drawing — 1-3 marks

Recognise it: "give the IUPAC name" or "draw the structure of...".

1. Longest chain, then check for a double bond — **it takes numbering priority over the halogen**.
2. Substituents alphabetically in the name; lowest locants where there is a tie.

Trap: numbering from the halogen out of habit when a double bond is present. 4-chlorobut-1-ene, not 1-chlorobut-3-ene.

5 · Conversions — 3 marks

Recognise it: "how can the following conversions be carried out".

1. Count carbons. Choose the route. Write reagents above every arrow — the reagent is what's being marked.

6 · Definitions and stereochemistry — 1-2 marks

Recognise it: "define the following term", or a pair of structures with "which is chiral?".

1. For chirality: find the carbon with **four different** groups. Two identical groups anywhere on it and it is not chiral.
2. For definitions, give the definition and **one example** — the example is usually worth half the mark.

Past year questions · mark slots — *what each type is worth*

Slot	What turns up there
1 mark	Name the type of halide (vinylic / allylic / benzylic). One-line reason questions. Assertion-reason on reactivity orders. (<i>inferred</i>)
2 marks	"Give the reason" pairs. Two differences between S_N1 and S_N2 . Arrange in order of boiling point. One-step "what happens when".
3 marks	IUPAC naming (three compounds). Three-part "give reason for the following". Conversions. Identify A and B in a two-step scheme.
5 marks	Multi-part sets: a reactivity comparison plus a halide-type identification plus a polyhalogen recall, or a product-prediction set combined with definitions.

The video groups these as short answer type I (2 marks, Q1-10), short answer type II (3 marks, Q11-23) and long answer (5 marks, Q24-25). At 6 marks in your blueprint this chapter will most likely appear as a mix of 1- and 2-markers rather than a 5-marker — but the 5-mark sets are still the best revision, because they bundle four separate ideas into one question.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2021 Q7 · 2024 Q13(b) · 2026 Q24(a)

Which of a pair reacts faster by S_N2 , with reason. Runs across RI vs RCl (leaving group), tertiary vs primary (steric hindrance), and two primaries differing in where the branch sits. The single most repeated question shape in the chapter.

2026 Q13(a) · 2024 Q22(a)

KCN gives cyanide but AgCN gives isocyanide — the ambident nucleophile question. Ionic versus covalent is the whole answer.

2020 Q4 · 2026 Q10(a) · 2021 Q21

S_N1 : write the mechanism, name the most reactive isomer, or give two differences from S_N2 . Carbocation stability is the thread through all three.

2017 Q5(a) · lecture PYQ

Why alkyl halides, though polar, are immiscible with water. Energy of bonds broken versus energy released.

2023 Q8(a) · 2024 Q11(c) · lecture PYQ

Arrange in increasing order of boiling point. Appears in both a chain-length form and a halogen-count form.

2024 Q11(b) · lecture PYQ

Why thionyl chloride is preferred for making alkyl chlorides from alcohols. Escapable gaseous by-products.

2026 Q10(b) · 2020 Q18

Wurtz reaction run backwards — the product is given, find the starting halide. Cut the symmetrical alkane in half.

2024 Q13(c) · lecture PYQ

Why Grignard reagents must be prepared under anhydrous conditions. Water destroys them, giving the alkane.

2025 Q19(b) · 2026 Q25(b) · lecture PYQ

Friedel-Crafts on a haloarene, alkylation or acylation, asking for the **major** product. Always para, always because of symmetry and less crowding.

2025 Q15 · 2026 Q23(b) · 2020 Q18

Multi-step "identify A, B, C, D, E" schemes built from Grignard, Wurtz, alcoholic KOH and HBr addition. Three separate papers, near-identical shape.

Past year questions · cold practice — *answers only* — *work them before you look*

Reactivity and mechanism

- 2021 Q7 — which shows faster S_N2 : *n*-propyl iodide or *n*-propyl chloride? The iodide — I^- is a better leaving group because of its large size and weak C-I bond
- 2021 Q7 — faster S_N2 : *tert*-butyl chloride or ethyl chloride? Ethyl chloride — primary, so far less steric hindrance
- 2026 Q13(b) — why is cyclohexylmethyl chloride more reactive to S_N2 than cyclohexyl chloride? The first is primary, the second secondary — less crowding at a primary carbon
- 2026 Q10(a) — most reactive C_4H_9Br isomer towards S_N1 . 2-bromo-2-methylpropane — it forms the most stable, tertiary carbocation
- 2021 Q21 — faster S_N1 : cyclohexylmethyl chloride or benzyl chloride? Benzyl chloride — the benzyl carbocation is resonance-stabilised by the ring
- 2026 Q24(a) — more reactive to S_N1 : 2-bromo-2-methylbutane or 1-bromopentane? 2-bromo-2-methylbutane — tertiary carbocation beats primary
- 2025 Q20(b) — why is allyl chloride hydrolysed faster than *n*-propyl chloride? The allyl carbocation is resonance-stabilised; the propyl one is not

- 2026 Q23(c) — which of S_N1/S_N2 gives inversion, which gives racemisation? S_N2 inversion (back-side attack); S_N1 racemisation (planar carbocation, attack from either face)

Reason questions

- Lecture PYQ Q1 — why are haloarenes less reactive to nucleophilic substitution? Resonance gives the C-X bond partial double-bond character, and the sp^2 carbon holds the pair more tightly
- 2024 Q16 — why is the dipole moment of chlorobenzene lower than that of cyclohexyl chloride? The sp^2 carbon is more electronegative, so less charge separation; resonance also shortens the C-Cl bond
- 2020 Q9(b) — why does $-NO_2$ at ortho/para increase haloarene reactivity to nucleophilic substitution? It withdraws electron density, weakening the C-X bond and activating the ring
- 2026 Q25(a) — chlorine shows $-I$ yet is ortho/para directing. Why? $-I$ deactivates the whole ring, but $+R$ resonance raises electron density specifically at ortho and para
- 2024 Q22(c) — why are d- and l-butan-2-ol hard to separate by fractional distillation? Enantiomers have identical boiling points
- 2026 Q24(a) — why is chloroform stored in dark bottles, filled to the brim? Light and air oxidise it to phosgene, $COCl_2$, a poisonous gas

Products and conversions

- 2023 Q3 — isopropyl alcohol + PCl_5 , then AgCN. Isopropyl chloride, then isopropyl isocyanide ($(CH_3)_2CHNC$)
- 2023 Q3 — 1-chloropropane + alcoholic KOH, then HBr. Propene, then 2-bromopropane (Markovnikov)
- 2020 Q9(a) — convert but-1-ene to 1-iodobutane. HBr/peroxide $\rightarrow$ 1-bromobutane, then NaI /acetone (Finkelstein)
- 2025 Q12(a) — convert chlorobenzene to biphenyl. Na/dry ether — Fittig reaction
- 2025 Q12(c) — convert 2-bromobutane to but-2-ene. Alcoholic KOH; but-2-ene is major by Saytzeff
- 2025 Q20(a) — methyl bromide + AgF. Methyl fluoride — Swarts reaction
- 2025 Q20(b) — 2,4,6-trinitrochlorobenzene, hydrolysed. 2,4,6-trinitrophenol, i.e. picric acid
- 2026 Q25(a) — chlorobenzene + conc. H_2SO_4 , heat: major product. 4-chlorobenzenesulphonic acid
- 2026 Q25(b) — bromobenzene + CH_3Cl / anhydrous $AlCl_3$: major product. 1-bromo-4-methylbenzene (para major)
- 2025 Q19(a) — a nitro-substituted ethylbenzene heated with Br_2 : major product. Bromination at the benzylic carbon — that free radical is resonance-stabilised
- 2026 Q25(b) — dehydrohalogenation of 1-bromo-1-methylcyclohexane. 1-methylcyclohexene — the more substituted alkene, by Saytzeff

Naming, structure and identification

- 2026 Q2(a) — draw 4-bromo-3-methylpent-2-ene. $CH_3-CH=C(CH_3)-CHBr-CH_3$

- 2024 Q11(a) — IUPAC name of a four-carbon chain with a double bond and Cl at the far end. 4-chlorobut-1-ene — the double bond takes numbering priority
- 2017 Q14 — IUPAC names of $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$, 1,3-dibromobenzene, and a chloropropene. 2-bromobutane; 1,3-dibromobenzene; 3-chloroprop-1-ene
- 2026 Q23(a) — a hydrocarbon C_5H_{12} gives only one monochlorination product. Identify it. 2,2-dimethylpropane — all twelve hydrogens are equivalent
- 2026 Q10(b) — which $\text{C}_4\text{H}_9\text{Br}$ isomer gives 2,5-dimethylhexane with Na/dry ether? 1-bromo-2-methylpropane — cut the product in half
- 2026 Q24(a) — halogen present in $\text{CH}_2=\text{CH}-\text{Cl}$? Vinylic — chlorine sits directly on a doubly-bonded carbon
- 2017 Q5(b) — which of the pair is the allylic halide? The one where Cl is on an sp^3 carbon *next to* the double bond, not on it
- 2026 Q17(b) — which of a pair is chiral? The one whose carbon carries four *different* groups; two identical groups anywhere and it isn't
- 2026 Q24(b) — define ambident nucleophile and racemic mixture. Two donor atoms, attacks through either (CN^- , NO_2^-); and a 1:1 mixture of enantiomers, optically inactive because the rotations cancel

Built from Sourabh Raina's Haloalkanes and Haloarenes one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 6 (Rationalised 2022-23). NCERT spellings used throughout: **Zaitsev** (NCERT notes it is "also pronounced as Saytzeff"), **ambident** nucleophile, **vicinal** and **geminal** dihalides, **laevorotatory**. C-Cl bond lengths (177 pm in haloalkanes, 169 pm in haloarenes) are NCERT's. First-contact material: every term marked **[exposure]** is explained from scratch before use.

Two stretches of both transcripts were lost at automatic-transcription chunk boundaries and were recovered by re-transcribing those windows from the source audio: the S_N2 mechanism and ambident-nucleophile block in the lecture, and PYQ questions 12, 13, 17, 18, 22 and 23. All twenty-five PYQ questions are accounted for here.